

CST8506 Assignment 1

Due: Feb 13, 2026, at 11:59 PM Sharp!!!

(Late submissions will not be accepted)

Goal: The goal of this lab is to explore and analyze **Wine Quality dataset** from **UCI Machine Learning Repository** (<https://archive.ics.uci.edu/dataset/186/wine+quality>) and then reduce the dimensionality by applying PCA and LDA approaches, and perform classification using kNN, Random Forest, SVM, Logistic Regression and MLP. Compare the results before and after dimensionality reduction. You need to follow CRISP-DM for this assignment.

Seeds dataset is tab-separated. When you download the file, if you have extra tabs in some rows, remove them manually or in python.

There is no need to create a professional report for this assignment. Just like we do in labs, include code for each step and explain it (make sure to explain parameters too. Even if you are using default parameters, make sure to specify them and explain) in a word document.

Data Understanding

Thoroughly analyze your data to have a clear understanding of your data and their attributes and types. Perform all CRISP-DM steps of this phase.

Data Preparation

Perform all steps in Data Preparation phase of CRISP-DM while preparing your data. Make sure to include attribute names. Print basic stats like number of instances, number of attributes, first few instances etc. As part of data formatting, reduce the dimensionality by applying PCA and LDA techniques. Save both new dataframes separately (X_train_scaled, X_train_pca, X_train_lda, X_test_pca, X_test_lda etc.).

Modeling & Evaluation

As you will be performing the same task - classification - for different datasets using different methods, write a function that takes the datasets and methods as parameters. The results should be added accordingly to a data structure.

You have 3 sets of data frames:

1. Standardized Dataset
2. After PCA
3. After LDA

For all three sets of data listed above, perform classification using the following methods (include three sets of parameters for each method):

1. Random Forest
2. MLP
3. SVM
4. Logistic Regression
5. kNN

Tabulate best results (Accuracy and F1 measure) for each dataset and each method **one** table (Make sure to include the best parameters also in the table – see the sample below).

Method	Parameters	Standardized	After PCA	After LDA
kNN				
RF				
SVM				
LR				
MLP				

Visualizations to be included:

1. scree plot, cumulative scree plot
2. After applying PCA, actual classes of the train set color coded by class and then in the second plot, support vectors should also be plotted, color coded by class (2 plots in parallel)
3. After applying LDA, actual classes of the train set color coded by class and then in the second plot, support vectors should also be plotted, color coded by class (2 plots in parallel)

Submission Details:

Assignment should have a **cover page**, table of contents, figures etc. Upload the answer document and notebook to Brightspace (Don't zip, zipped files will not be graded). There will be mark deduction if you are not following the submission requirements. As mentioned earlier, no need to spend time on creating a professional report.